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Cross-matching libraries

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Abstract

https://lamortadella.github.io/BPRPlibraries/index.html.

Key words: stars: evolution — stars: supergiants — stars: massive

1 Introduction

2 XSP and Gaia BP/RP

The work of Verro et al. (2022) delivers a list of 830 spectra, which are freely available on Vizier. A list with 807 matches (2arcsec radius) between the XSP spectra and Gaia DR3 data points was used xsl_dr3_xmatch_GaiaEDR3_xmatch_GaiaDist_KV_2asec_pure.csv.

The used table contains five entries for which no spectra are available in the tar file, and that are listed in Table 2; in other words these five entries are not included in the list of 830 published spectra (full table retrieved via Vizier).

A best quality list of 449 spectra was also received from Verro et al. with 447 belonging to the table xsl_dr3_xmatch_GaiaEDR3_xmatch_GaiaDist_KV_2asec_pure.csv. When crossmatching the Best XSP spectra with the table gaiadr3.astrophysical_parameters 447 entries (all) are found. The matches with has_xp_continuous=1 are 345. When considering the repeated XSP observations, the 345 matched corresponds to 292 stars.

In the github webpage https://lamortadella.github.io/BPRPlibraries/index.html, there is a display page with the original XSP spectra (cyan) and overlaid the Gaia BP/RP spectra (black). A rebinned and shifted version of the XSP spectra is created (red curve) to match the resolution and wavelength scale of the Gaia data – note that the accuracy of the latter can be improved removing the shifting and adding proper summation...

In Table 3, the 345 XSP spectra Ids and associated Gaia-DR3 Ids are listed. chiT is a measure of the quality of the spectral match and is defined as: chiT=$\sum \frac{flux(XSL)-flux(BPRP)}{flux(BPRP)}$. The Table 3 also lists a "factor" which is simply the ratio of the average flux(BP/RP) and average Flux(XSL) between 700 and 750 nm. In Fig. **??**, the histogram of the chiT values is seen to peak at 7.7. In Figs. **??** and **??**, the chiT values are plotted versus the G-mag and the BP-RP color.

All sources with a BP/RP spectrum have more than 15 observations in BP and RP bands (Figs. 9 and 10).

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3 UVES_POP

For the UVES_POP spectra, Names, coordinates, and Gaia IDs were read from the fits header. The reported Gaia IDs were a mix of Gaia DR2 IDs and Gaia DR3 IDs. The annotated target coordinates were up to 2″ off. By using the Names (mostly HD-names) coordinates were retrieved from Simbad. By cross-correlations, 2MASS-IDs, Gaia DR3 and Gaia DR2 IDs were re-retrieved. A few new matches were located, as listed in Table 4. The other matches were all fine (beside the DR2 and DR3 ambiguity).

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Library	Nlist	N_gaiaID	$N_{BP/RP}$
UVES_POP	406	376	302
XSL	447[a]	447	345

(a)= 447 Best quality are included among the list of 807 matches (2″ radius).

Table 1. Three matches found between the XSP spectra and those from the SPSS library.

Gaia-DR3 ID	XSP	SPSS	SPSS_ID
2633603478379307904	X0758	23	Feige110
2633603478379307904	X0766	23	Feige110
2633603478379307904	X0790	23	Feige110

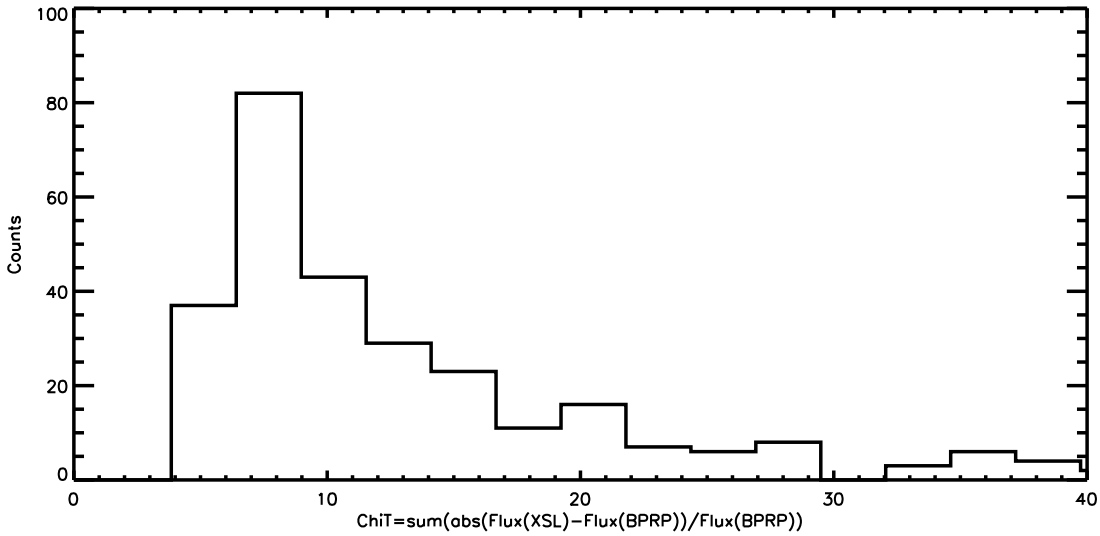

Fig. 1. XSL spectra: Histogram of the ChiT parameter.

Table 2. Xshooter IDs not included in the published list of 830 entries

X0008
X0227
X0429
X0457
X0833

3.1 UVES_POP Gaia BP/RP

There are 406 fits available in the UVES_POP library (Borisov et al. 2022). 376 have a Gaia DR3 match and 302 Gaia DR3 matches with has_xp_continuous=true are found in the table gaiadr3.astrophysical_parameters. A number of 302 BP/RP mean spectra in the continuous representation could be retrieved and resampled.

UVES_POP spectra are resampled to the Gaia BP/RP resolution, and compared with the Gaia BP/RP spectra. A ChiT parameter is calculated as the $\sum \frac{flux(XSL)-flux(BPRP)}{flux(BPRP)}$.. The histogram of the obtained ChiT is shown in Fig. 6. It results the 101 stars are suitable for a calibration study (ChiT < max(chiT)=7.69). Only 32 stars have a ChiT smaller than five. The Table 3 also lists a "factor" which is simply the ratio of the average flux(BP/RP) and average Flux(XSL) between 700 and 750 nm.

In Fig. 6, the histogram of the chiT values is seen to peak at 7.69. In Figs. **??** and **??**, the chiT values are plotted versus the G-mag and the BP-RP color. The variables are shown in red, they have a similar distribution as he non variable.

All sources with a BP/RP spectrum have more than 15 observations in BP and RP bands (Figs. 9 and 10).

section**References**

Borisov, S., Chilingarian, I., Rubtsov, E., et al. 2022, arXiv e-prints, arXiv:2211.09130

Verro, K., Trager, S. C., Peletier, R. F., et al. 2022, A&A, 660, A34

Table 3. 345 matches found between the Best-selected XSP spectra and those from the SPSS library.

Gaia ID	XSP	chiT	factor	G [mag]	BP-RP [mag]
2804091414920169344	X0007	12.24	1.03	8.388	0.989
2527043140743265664	X0024	220.07	3.88	9.937	0.314
5171417602530766976	X0065	5.73	0.99	9.643	0.650
52624073910834176	X0072	5.36	1.00	9.093	0.657
3282524177258336512	X0074	11.33	1.05	5.666	0.103
219057458247921536	X0075	15.40	1.05	8.232	1.107
2593188145361730432	X0077	11.86	1.04	7.969	1.292
4646981136249320832	X0089	19.56	1.07	6.395	0.902
2593188145361730432	X0092	21.14	1.06	7.969	1.292
2993364748147267456	X0122	30.20	1.09	7.783	1.111
6734732303270177024	X0137	111.83	1.52	14.402	1.530
6734782399770915200	X0138	22.89	0.93	14.254	1.358
4387079367952007168	X0140	95.26	1.46	6.806	0.769
4494267870326116864	X0141	126.22	1.72	5.976	0.762
4164893700353983104	X0143	47.58	1.19	9.435	0.977
6734635821123930368	X0151	18.73	1.04	14.236	1.198
4050203741234555392	X0152	23.02	1.04	15.386	1.935
4165370682239910144	X0157	26.12	1.09	10.352	0.888
4285492327595742080	X0162	50.26	1.18	7.300	1.776
4279028985902356096	X0164	44.74	1.16	7.662	0.360
4050202886598183168	X0170	29.70	1.08	14.378	1.940
6734635821123930368	X0171	11.06	1.02	14.236	1.198
1850440690450639232	X0175	14.39	0.97	7.998	0.589
1800518563283501440	X0176	8.51	0.98	9.924	0.695
6916198691587462272	X0177	8.63	0.98	7.747	0.995
2688034762857131136	X0178	19.16	0.97	6.426	0.226
2699301565882761344	X0179	5.87	0.99	8.345	0.769
6905364821265383424	X0195	10.62	0.98	6.048	1.095
4246840611198113024	X0196	13.32	0.98	6.743	0.001
1841916486040590080	X0197	9.29	1.01	7.832	1.217
1848170611255790848	X0198	9.09	0.97	10.361	0.751
4231133434759619200	X0199	16.74	0.97	6.195	-0.105
6858374206854249728	X0200	7.14	0.99	8.096	0.703
4651961099314703232	X0206	29.90	0.92	16.460	1.623
4308878561939677824	X0207	13.19	0.97	6.679	1.414
1820374029621976704	X0211	7.79	0.99	7.422	1.121
1833199729671740800	X0212	9.63	0.98	7.036	1.045
1833577819925426944	X0213	7.26	0.98	8.739	0.741
4279028985902356096	X0220	14.78	0.96	7.662	0.360
4268663477469514496	X0222	13.71	0.97	7.529	0.685
2658240166703766016	X0223	6.88	0.98	10.084	0.679
4565784161044390656	X0225	40.73	1.13	11.099	0.779
4474806789394684544	X0228	13.61	1.00	6.749	0.190
4651957216664541184	X0238	15.05	0.98	16.533	1.633
1172688357148006912	X0243	12.02	1.05	8.720	-0.025
4358758216162768000	X0245	11.76	1.05	6.216	0.122
1833199729671740800	X0249	12.03	1.04	7.036	1.045
1820374029621976704	X0268	15.67	1.05	7.422	1.121
4468813931327385600	X0269	22.52	1.09	6.596	0.869
4474747621924408320	X0270	28.66	1.16	6.256	0.037

Table 3. Continuation of Table 3 .

Gaia ID	XSP	chiT	factor	G [mag]	BP-RP [mag]
4308878561939677824	X0272	19.11	1.06	6.679	1.414
1841916486040590080	X0274	54.73	1.21	7.832	1.217
2699301565882761344	X0275	53.59	1.21	8.345	0.769
1850440690450639232	X0276	101.99	1.48	7.998	0.589
4279028985902356096	X0283	35.24	1.13	7.662	0.360
3320346999297121536	X0285	91.51	1.48	7.148	0.230
5504529287416164992	X0287	8.26	1.02	6.789	0.644
4246840611198113024	X0288	10.19	1.04	6.743	0.001
4231133434759619200	X0289	9.76	1.02	6.195	-0.105
6858374206854249728	X0290	10.70	1.04	8.096	0.703
2688034762857131136	X0291	11.12	1.02	6.426	0.226
6916198691587462272	X0292	10.39	1.00	7.747	0.995
4268663477469514496	X0295	12.91	1.05	7.529	0.685
6905364821265383424	X0297	8.95	1.01	6.048	1.095
49415527182759296	X0313	9.61	1.01	5.481	-0.091
5676959889203060096	X0347	39.20	1.12	7.981	1.315
6083695761740022144	X0351	13.96	0.96	13.281	1.378
6045460111755316736	X0352	8.45	1.02	11.747	1.625
6045465987271055232	X0354	11.60	1.03	11.888	1.786
5949553973093167104	X0355	25.68	1.09	10.535	1.369
4331831966671986560	X0356	16.85	1.05	10.075	1.217
5805105976563238144	X0357	12.36	1.07	9.310	-0.216
3117566200092213504	X0358	7.46	1.02	8.283	0.443
3939515154043771392	X0359	10.66	1.04	12.041	0.673
3654035240588664960	X0360	172.45	2.14	9.161	1.554
3674725369363742592	X0361	234.66	4.23	10.627	0.906
3674612772501124608	X0363	246.25	4.47	9.111	1.220
3507882897984968320	X0371	17.27	1.08	6.983	0.384
6301027158526567040	X0372	6.91	1.00	6.896	1.276
3646061719703116032	X0373	10.12	1.02	6.861	0.908
6307374845312759552	X0374	11.56	1.03	8.828	1.012
3941562272896500992	X0375	61.06	1.24	8.528	0.978
3732696398384550400	X0376	117.18	1.57	8.564	0.825
3636549599797661824	X0378	9.80	1.00	10.458	1.293
3725115471868587904	X0379	11.88	1.02	8.151	1.260
4421479204133344128	X0380	36.78	1.12	11.948	1.629
4421477829743733248	X0381	84.21	1.34	12.171	1.561
6307365499463905536	X0383	28.15	1.11	9.156	1.107
3650718804281798912	X0384	8.36	1.00	7.849	0.942
4551981991583160704	X0387	56.85	1.25	9.723	0.636
1745936164681118592	X0388	22.25	0.94	13.758	1.314
1821609021711481728	X0389	22.81	1.07	14.154	1.313
1235563517265388928	X0392	64.49	1.25	9.601	1.000
1275876252107941888	X0393	172.94	2.14	8.728	1.458
1275876252107941888	X0394	73.78	1.30	8.728	1.458
1262035427936249728	X0395	54.37	1.19	7.543	1.135
4050199996022387072	X0400	143.19	1.82	14.696	1.726
4050199996022387072	X0401	100.58	1.46	14.696	1.726
4050198351112514560	X0405	17.84	1.03	14.483	1.900
1821608471955735296	X0407	6.02	1.00	12.311	1.829

Table 3. Continuation of Table 3 .

Gaia ID	XSP	chiT	factor	G [mag]	BP-RP [mag]
1821607608629456512	X0408	12.97	0.97	14.198	1.456
1283756555019878528	X0409	10.43	0.98	7.929	1.132
6451102215021927040	X0412	5.73	0.99	9.660	0.922
4049030566622522240	X0419	38.24	1.16	15.483	1.768
4685877597088703360	X0422	52.14	1.22	7.983	1.574
5084391257193086336	X0423	34.87	1.13	6.452	1.042
4049028883040962816	X0426	55.48	0.85	15.495	1.672
4048839320356942720	X0427	59.95	0.83	15.587	1.927
6614285855300051840	X0430	12.43	1.02	7.299	1.279
2689931630869716096	X0433	10.38	1.02	6.634	0.217
2686137834717137792	X0435	15.79	1.01	9.379	1.271
1733244570676906880	X0436	9.82	1.00	6.745	1.200
1736838805468812160	X0437	9.46	1.00	7.817	1.496
1730456415347350784	X0438	9.71	1.02	7.064	1.364
2612247904551451264	X0441	8.97	0.97	9.296	1.147
2869759781250083200	X0442	7.51	0.98	7.300	1.379
2722849325377392384	X0445	6.38	0.99	10.245	0.693
2672702897842335872	X0446	8.14	0.97	7.994	1.208
2582106305105033216	X0448	8.68	1.07	10.263	-0.376
4158615798208505472	X0449	6.76	0.99	7.751	1.338
4469871906338470528	X0450	14.49	1.00	6.812	0.142
2537228466706526208	X0453	9.98	1.01	7.316	1.276
5949553973093167104	X0454	20.07	1.06	10.535	1.369
4435581505912059904	X0456	15.66	1.05	8.394	1.032
4364527594192166400	X0458	15.84	1.03	7.265	1.464
4471126479159876096	X0459	16.36	1.07	7.736	0.282
4470930186283476736	X0460	83.45	1.47	7.330	0.039
4478267193011680640	X0461	170.57	2.36	8.076	0.453
2582106305105033216	X0463	10.86	1.08	10.263	-0.376
4446275149846884352	X0464	7.39	1.02	9.039	0.940
4473687177317598336	X0465	14.71	1.05	9.323	0.421
4474760644265753984	X0467	8.72	1.01	6.314	-0.110
4476214061198753280	X0468	11.19	1.03	7.187	0.187
4468305755100841472	X0469	11.41	1.03	6.612	0.207
4284219883386710272	X0470	15.29	1.03	7.682	0.299
4286778069001143808	X0471	12.95	1.01	7.382	1.405
4286975912366995200	X0472	12.05	1.05	7.962	0.262
4268291017906171392	X0473	11.15	1.02	7.422	1.198
4292688321964796800	X0476	11.28	1.03	6.631	0.532
6858449008002621312	X0479	21.30	0.98	9.952	1.350
4247355045211154560	X0480	7.44	1.03	10.308	0.719
4273225145054713984	X0481	9.70	1.02	7.585	1.345
4240746602368593664	X0482	10.13	1.02	7.123	1.266
4243275719623780608	X0483	8.98	1.01	7.285	0.406
4243054649066947712	X0484	8.98	1.01	7.703	1.069
2699572561138691968	X0485	11.46	1.02	7.744	0.445
2681306546265329920	X0486	8.39	1.01	8.407	0.992
4500356450325206272	X0488	18.88	0.97	6.117	0.600
1833199729671740800	X0490	48.62	1.19	7.036	1.045
1821609021711485696	X0491	132.13	1.70	13.799	1.495

Table 3. Continuation of Table 3 .

Gaia ID	XSP	chiT	factor	G [mag]	BP-RP [mag]
1821620424811789568	X0494	74.57	1.29	12.266	1.684
6888560851915102464	X0496	29.47	1.10	9.949	1.091
4455529017661900160	X0500	8.02	1.00	8.741	1.405
4455529017661900160	X0501	8.53	1.00	8.741	1.405
6045465987271055232	X0502	12.61	1.03	11.888	1.786
4687246145432125824	X0504	6.57	0.99	13.043	1.570
4687250268600675712	X0506	43.99	1.14	12.692	1.693
5030366581563257088	X0508	56.52	1.20	9.370	1.386
4662061316228099584	X0518	25.47	1.09	11.290	0.737
3318883274444631296	X0520	10.12	1.03	8.194	1.084
2957939583009621120	X0521	16.03	0.96	12.519	1.686
2957939686088825472	X0522	9.61	0.99	13.022	1.542
2957940132765349888	X0523	10.19	1.00	12.741	1.642
4658168495332042112	X0524	9.91	0.99	14.722	1.515
2787924539542784896	X0526	10.07	1.02	7.049	1.244
2490426551679224064	X0529	9.25	0.98	7.067	0.684
3124766042749774720	X0535	14.68	1.00	7.772	0.316
3155993478647980416	X0536	16.39	0.96	7.667	1.095
3156126175957731200	X0537	15.19	0.96	7.192	1.261
3317535337611336832	X0539	9.36	1.03	6.998	-0.191
4660601611448785152	X0542	23.80	1.05	9.442	0.851
3261878613062758144	X0546	9.23	1.02	6.889	0.820
53848448829915776	X0547	40.88	1.15	7.987	0.960
5097726306014470016	X0548	8.32	0.98	7.099	1.368
3297303572040367232	X0549	5.85	1.00	8.580	1.564
5546929513812569856	X0550	20.51	1.06	7.458	1.640
3143312158208562048	X0552	15.97	1.05	7.617	1.378
884951489919636352	X0555	21.01	1.07	8.057	1.117
3311148828615843328	X0558	8.84	1.03	8.887	1.013
2500840472982387200	X0559	17.11	1.04	7.072	1.177
3245657582441858688	X0561	13.55	1.03	9.306	1.025
3145978027229218176	X0567	21.62	0.94	7.923	1.166
604911375882674560	X0568	23.55	0.93	9.258	1.519
628852004626257152	X0569	134.96	1.81	5.895	0.721
2984998873403435008	X0570	16.90	0.96	9.219	1.168
3374633977170787072	X0571	18.47	0.95	9.145	0.840
2999846575346395264	X0572	14.46	0.95	12.658	1.277
5648333210624292224	X0573	8.45	0.98	8.815	0.596
3122668346360478976	X0578	7.43	0.99	6.799	1.099
3122130169776904832	X0579	8.69	0.99	7.347	0.606
3316978400610184832	X0581	7.37	0.99	6.301	1.183
3328169371696096384	X0582	10.17	0.99	7.022	-0.110
3124790609962295552	X0583	7.91	1.00	7.129	1.149
5617564477239812864	X0585	6.71	1.00	7.026	0.792
3111514591172555520	X0586	9.85	0.98	7.674	1.087
3142916535882924416	X0588	9.25	0.99	7.459	1.624
2889759622882091392	X0593	9.37	1.02	8.226	1.230
4022846242663965952	X0594	7.80	1.02	8.393	1.320
3796826582305099520	X0595	9.93	1.01	8.875	1.004
3473922557216744448	X0598	8.19	1.00	8.809	1.478

Table 3. Continuation of Table 3 .

Gaia ID	XSP	chiT	factor	G [mag]	BP-RP [mag]
3473240378970537088	X0604	6.24	1.01	10.017	1.056
3920615854833203328	X0607	17.83	0.95	10.021	0.693
6083695761740022144	X0610	25.50	0.93	13.281	1.378
3699954179060804864	X0611	28.07	0.91	9.526	0.656
3699954179060804864	X0613	6.39	1.02	9.526	0.656
6149244697914619392	X0614	8.94	1.01	9.048	0.053
1239044708517928064	X0617	11.23	1.02	5.362	0.310
1262035427936249728	X0618	10.09	1.01	7.543	1.135
4455529017661900160	X0620	13.60	1.04	8.741	1.405
6045460111755316736	X0621	36.46	1.11	11.747	1.625
3568795670365205888	X0622	5.76	1.01	13.637	0.757
3572694538597881600	X0623	6.65	1.01	14.173	0.664
3594472015653451264	X0624	8.06	0.99	14.988	0.608
3630092241022731136	X0625	17.32	1.03	8.737	1.498
3603035836843409408	X0626	9.32	1.02	9.039	0.124
3603949359207346048	X0627	8.91	1.01	8.430	1.519
3472600458906843392	X0630	6.40	1.02	12.035	0.901
3632963817501130240	X0631	5.61	1.00	13.991	0.636
1197801408889764480	X0633	13.40	1.01	5.769	0.268
4455529017661900160	X0634	8.19	1.00	8.741	1.405
4426498680874318592	X0636	10.06	1.01	8.505	0.555
4582148604918575744	X0639	9.45	1.02	6.923	0.302
6045502305516138368	X0640	60.02	0.88	13.396	1.246
4479121685351502208	X0641	29.67	1.11	7.189	1.311
4203240852063111424	X0646	10.49	1.04	6.933	0.613
4273885436158791168	X0648	13.28	1.05	7.881	0.324
4279979582440702720	X0654	11.84	1.05	7.877	0.473
4479434290260290304	X0655	16.74	1.07	6.877	0.394
4285211505458307840	X0656	12.55	1.05	7.967	0.197
4285866230243966208	X0658	16.64	1.05	7.571	1.316
6045465987271055232	X0661	81.44	1.32	11.888	1.786
4258375617747092864	X0663	150.54	1.85	8.270	1.607
4253988020990913408	X0664	112.02	1.55	7.230	1.214
4254196722022452224	X0667	101.69	1.44	7.408	1.486
4209904751452473984	X0668	26.46	1.16	7.399	0.476
4189688954569244032	X0669	89.07	1.38	7.279	1.521
6871465473332110464	X0670	69.83	1.29	8.809	1.298
4225764240307248768	X0673	86.25	1.44	7.641	0.415
4123849927791444480	X0674	24.43	1.15	8.269	0.668
4518081093181566080	X0676	10.04	1.04	6.493	-0.029
4474780435475367424	X0677	10.61	1.04	7.096	0.625
4283355976484990976	X0678	12.42	1.03	7.267	1.378
4290915668695014016	X0679	9.78	1.04	7.472	0.663
4283938992538977280	X0680	13.10	1.04	6.150	0.996
1821607608629456512	X0681	8.11	1.01	14.198	1.456
6795038076486104320	X0682	8.91	1.03	9.816	0.644
4354377620100892416	X0692	22.60	1.15	8.895	-0.436
4232490506987506432	X0695	29.35	1.14	7.719	0.241
4233114892151395712	X0696	62.23	1.27	7.295	0.666
4303364682926575744	X0697	75.70	1.33	8.647	0.835

Table 3. Continuation of Table 3 .

Gaia ID	XSP	chiT	factor	G [mag]	BP-RP [mag]
6734640322249801728	X0698	71.80	0.82	15.011	1.625
6795730493933072128	X0699	60.08	0.85	13.712	1.207
2666364625624238080	X0700	18.31	1.06	10.076	1.239
1821608471955735296	X0702	14.11	1.03	12.311	1.829
1821608437595882496	X0703	8.34	1.01	12.890	1.492
1821609021711485696	X0705	6.48	0.99	13.799	1.495
1821609223559201792	X0712	42.77	1.14	12.425	1.735
1821609090431005696	X0713	231.44	3.51	13.130	1.495
1821609090431005696	X0714	281.38	7.71	13.130	1.495
1821608639421563136	X0718	127.42	1.64	12.522	1.707
1821608639421563136	X0719	198.23	2.57	12.522	1.707
1821608639421563136	X0720	177.60	2.19	12.522	1.707
6833229746017792128	X0721	151.36	1.89	9.798	1.190
6833229746017792128	X0722	95.86	1.43	9.798	1.190
6833229746017792128	X0723	74.02	1.30	9.798	1.190
2699538613717163136	X0727	24.59	1.10	7.920	0.311
2699538613717163136	X0728	36.64	1.15	7.920	0.311
2699538613717163136	X0729	35.61	1.15	7.920	0.311
1792900051510535296	X0730	37.40	1.15	8.598	0.110
1792900051510535296	X0731	15.20	1.04	8.598	0.110
1792900051510535296	X0732	68.28	1.31	8.598	0.110
5033960575837265920	X0733	9.89	1.03	7.820	1.148
1821608708141030784	X0734	15.80	1.04	13.045	1.645
1821608708141023488	X0735	10.80	1.00	14.419	1.418
6858449008002621312	X0738	46.28	1.15	9.952	1.350
1821620463505562880	X0741	6.55	0.99	14.501	1.438
2806529032199025536	X0747	15.06	1.08	6.037	0.329
1821620424811789568	X0748	28.54	1.11	12.266	1.684
2686137834717137792	X0750	22.40	1.09	9.379	1.271
2324002513193850752	X0752	20.76	1.08	8.464	1.316
2788908774248323072	X0753	22.02	1.09	8.553	0.982
4688993681762966400	X0755	20.01	1.03	13.591	1.281
4688993402542368512	X0757	10.20	1.04	12.503	0.289
5142929531133360768	X0760	9.56	0.98	15.412	1.134
6858449008002621312	X0764	21.63	0.96	9.952	1.350
5040300909638155520	X0768	6.72	1.02	8.688	1.071
5040300909638155520	X0769	6.78	1.02	8.688	1.071
3219618436972965248	X0771	14.30	1.01	7.207	-0.175
3232901636947366656	X0772	7.78	1.00	8.550	1.231
5142632388116286336	X0778	8.58	1.03	8.779	0.854
102082477749475200	X0793	10.35	0.99	6.979	0.670
64929391436043264	X0794	12.20	1.03	8.094	0.273
3179824465465177984	X0796	193.29	2.88	6.154	0.624
4657660795812170368	X0797	190.62	2.34	12.127	1.928
4657766245829931136	X0809	72.71	1.00	15.852	2.208
3219964886216144640	X0816	8.90	1.03	6.990	0.535
3219618436972965248	X0817	10.16	1.03	7.207	-0.175
4657140039643647360	X0821	311.64	0.48	15.578	2.161
4657379389531307008	X0824	10.78	1.02	11.254	1.301
3325908775789740160	X0826	9.32	1.00	7.322	1.508

Table 3. Continuation of Table 3 .

Gaia ID	XSP	chiT	factor	G [mag]	BP-RP [mag]
4809142471233678208	X0827	9.67	1.01	7.927	1.470
2927005918574498304	X0829	9.75	1.02	7.090	1.344
3320296249963637504	X0831	12.19	1.02	7.824	0.455
3218445979621236992	X0832	8.08	1.02	7.490	0.791
3318989239876899328	X0834	15.81	0.99	7.718	0.139
3143033088413584256	X0840	17.02	0.96	7.040	1.248
3034578464047878272	X0841	17.30	0.95	8.927	1.284
3085071164726715648	X0843	8.97	1.02	8.999	1.052
3169175179954291968	X0846	6.33	1.00	8.792	0.721
3144388305214624896	X0847	8.25	1.02	7.963	0.306
3114203545639392000	X0851	17.98	0.95	7.533	1.042
3085071164726715648	X0853	14.04	0.96	8.999	1.052
3430863585931719680	X0854	24.96	0.93	6.911	1.300
3137401591716802432	X0855	46.74	0.88	7.612	1.048
595977294151338624	X0856	92.86	1.38	7.092	0.282
595977294151338624	X0857	9.45	0.99	7.092	0.282
3092651163524103424	X0858	14.57	1.04	7.642	1.173
604911375882674560	X0859	21.51	1.05	9.258	1.519
5525081496202448000	X0861	20.78	1.08	7.200	1.460
5709390701922940416	X0863	7.52	0.98	9.534	0.635
5660878702718068864	X0866	12.02	0.98	8.986	1.422
615943806835727872	X0867	6.96	0.99	8.207	0.606
866863321051682176	X0868	6.25	0.99	10.654	0.611
690419295663797760	X0870	56.91	1.25	9.231	0.475
661324431287688448	X0871	180.12	2.23	6.152	1.100
588856788129452160	X0872	48.09	1.16	11.015	0.609
581201571564630784	X0873	25.92	1.07	7.786	0.942
608373012147745920	X0874	36.17	1.12	8.692	0.077
685042030968334592	X0875	52.76	1.17	7.485	0.821
685042030968334592	X0877	8.99	0.97	7.485	0.821
646645946761688704	X0878	11.16	1.02	7.588	1.565
3890172886119687680	X0880	8.73	0.99	8.129	1.666
3989999608389935360	X0881	38.87	1.14	8.496	0.941
731253779217024640	X0882	10.41	1.03	10.102	0.749
735507548802653056	X0883	11.34	0.98	10.023	1.358
3896561220475435520	X0884	7.93	1.01	12.104	1.306
3871997546595984000	X0885	8.99	1.02	8.760	0.161
3546664150566178560	X0886	12.36	1.03	9.408	1.068
3520548825260557312	X0891	65.88	0.82	9.116	1.235
3710786979933623424	X0897	42.77	0.88	7.874	1.448
3501202540212450432	X0899	15.18	0.99	9.134	0.556
3702094687679734272	X0900	8.88	1.03	8.981	0.854
3702148666829533952	X0901	7.63	1.00	7.949	0.886
3703673070982634624	X0902	6.89	1.00	10.129	1.264
3525736595998110464	X0904	7.03	1.01	7.733	1.048

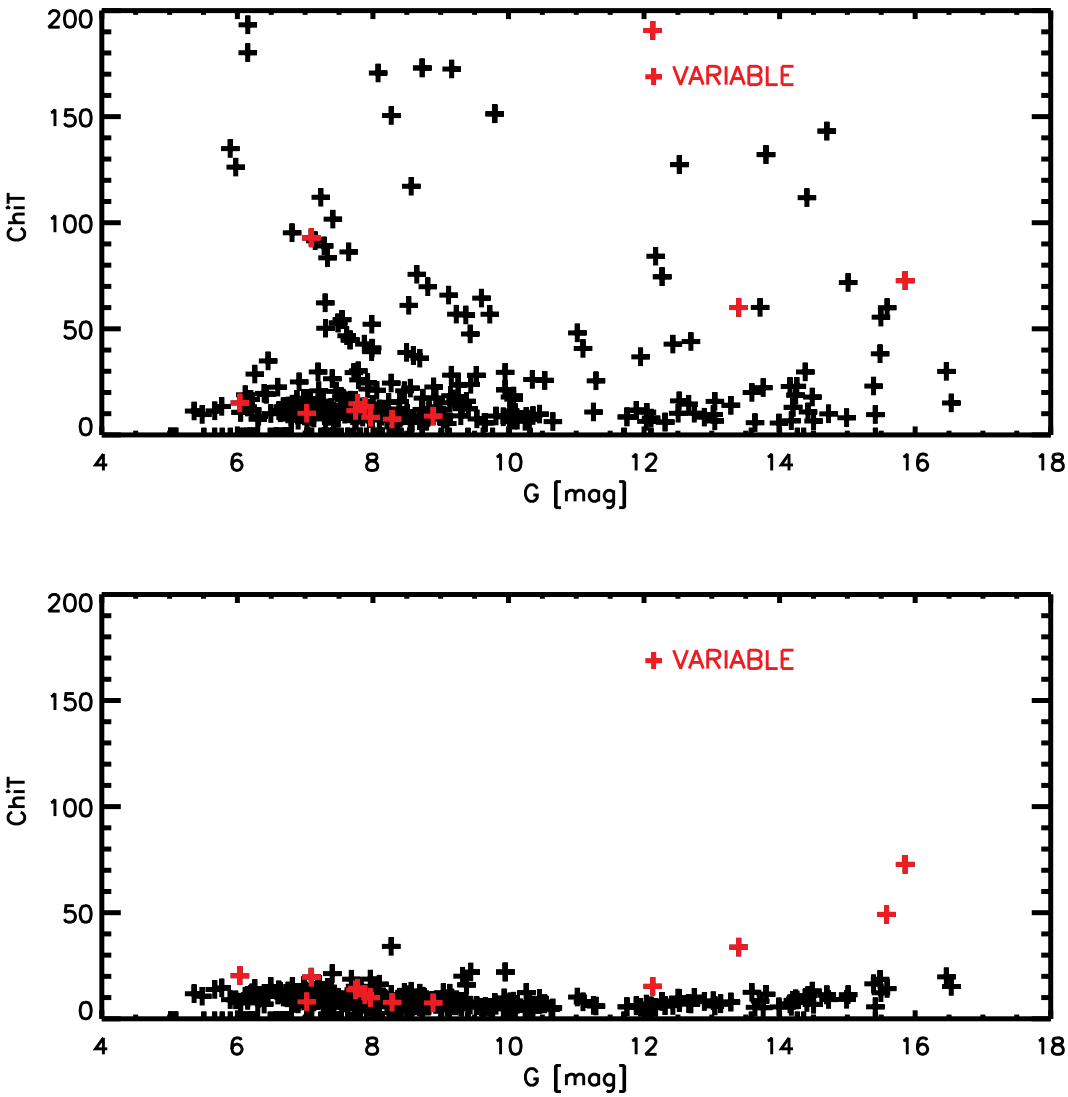

Fig. 2. *Upper panel:* ChiT versus Gmag of the XSL stars. *Lower panel:* An adjusted ChiT, i.e. a ChiT runned after a small rescaling of the XSL spectrum.

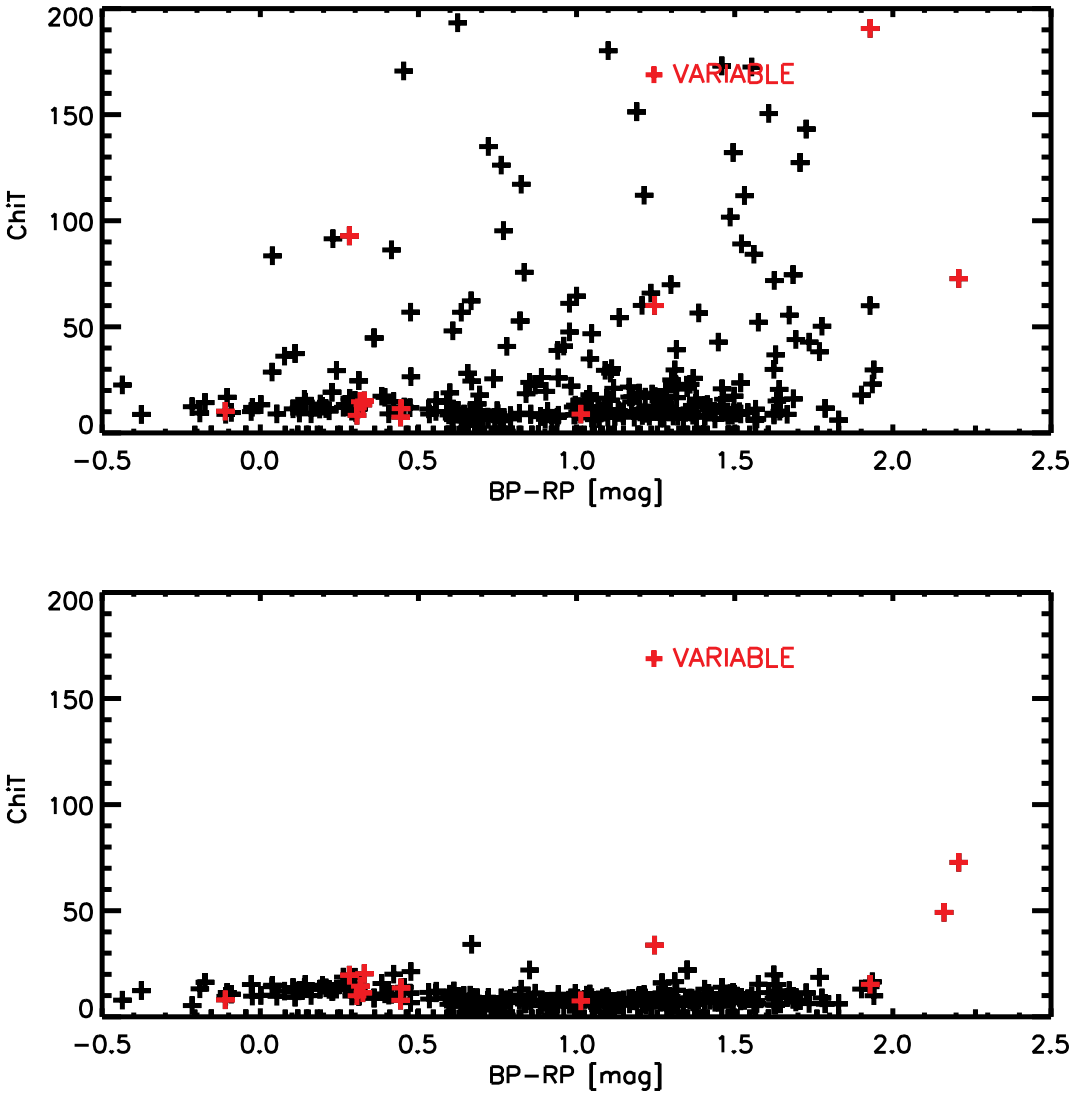

Fig. 3. *Upper panel:* ChiT versus BP-RP mag of the XSL stars. *Lower panel:* An adjusted ChiT, i.e. a ChiT runned after a small rescaling of the XSL spectrum.

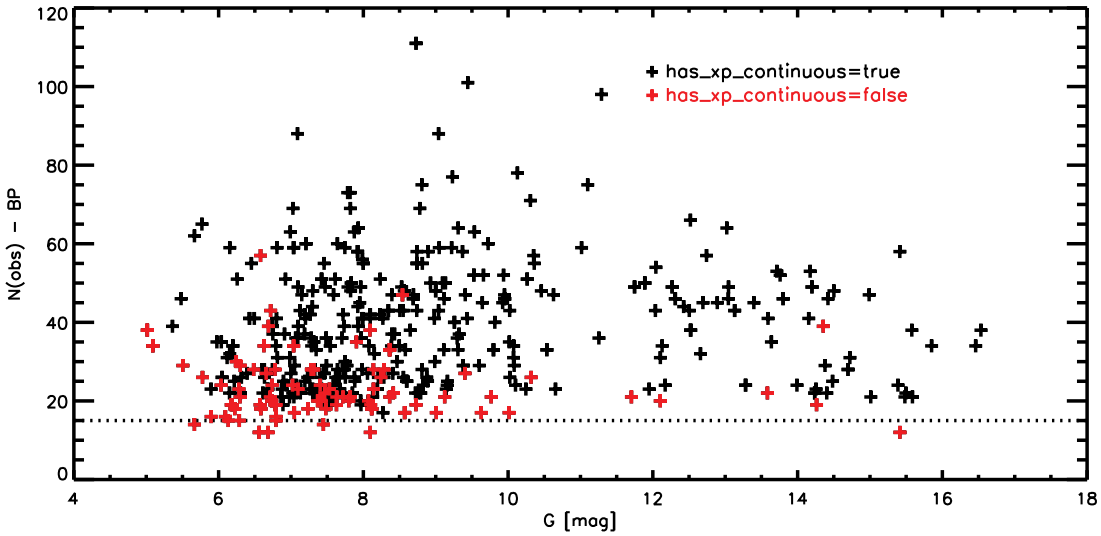

Fig. 4. XSL stars: Number of observation in BP-band versus Gmag.

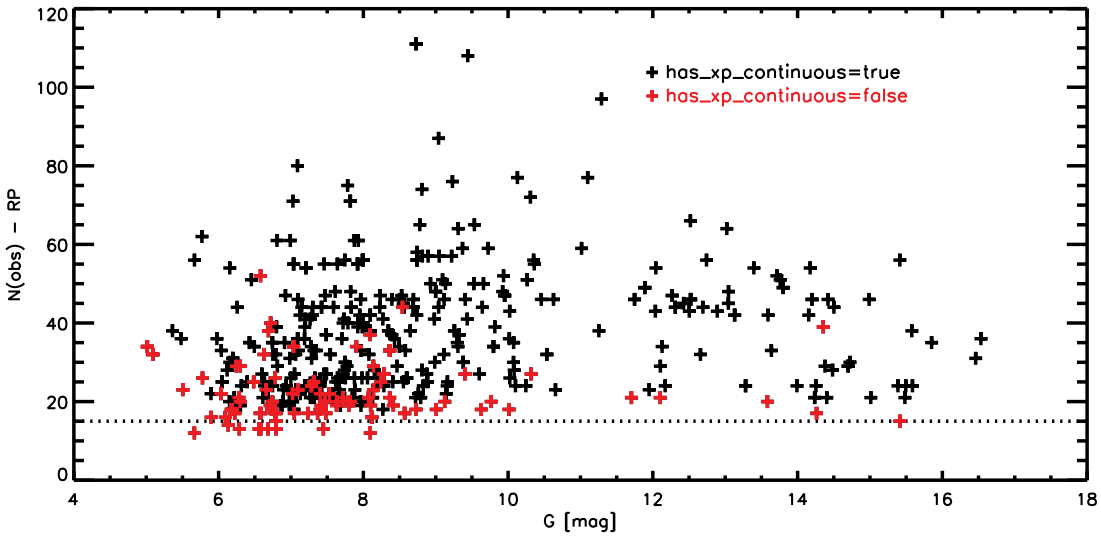

Fig. 5. XSL stars: Number of observation in RP-band versus Gmag.

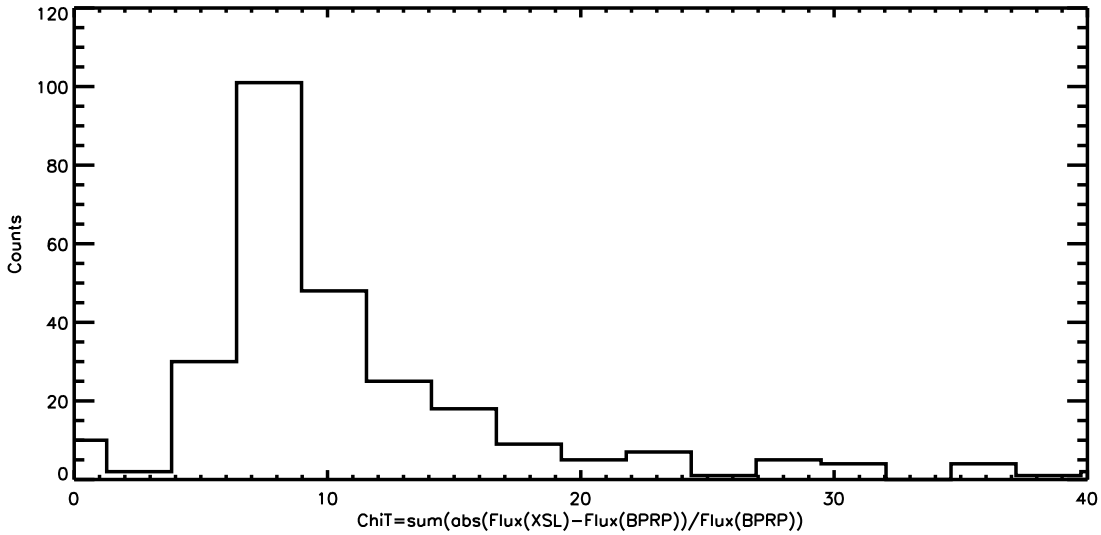

Fig. 6. UVES_POP spectra: Histogram of the ChiT parameter.

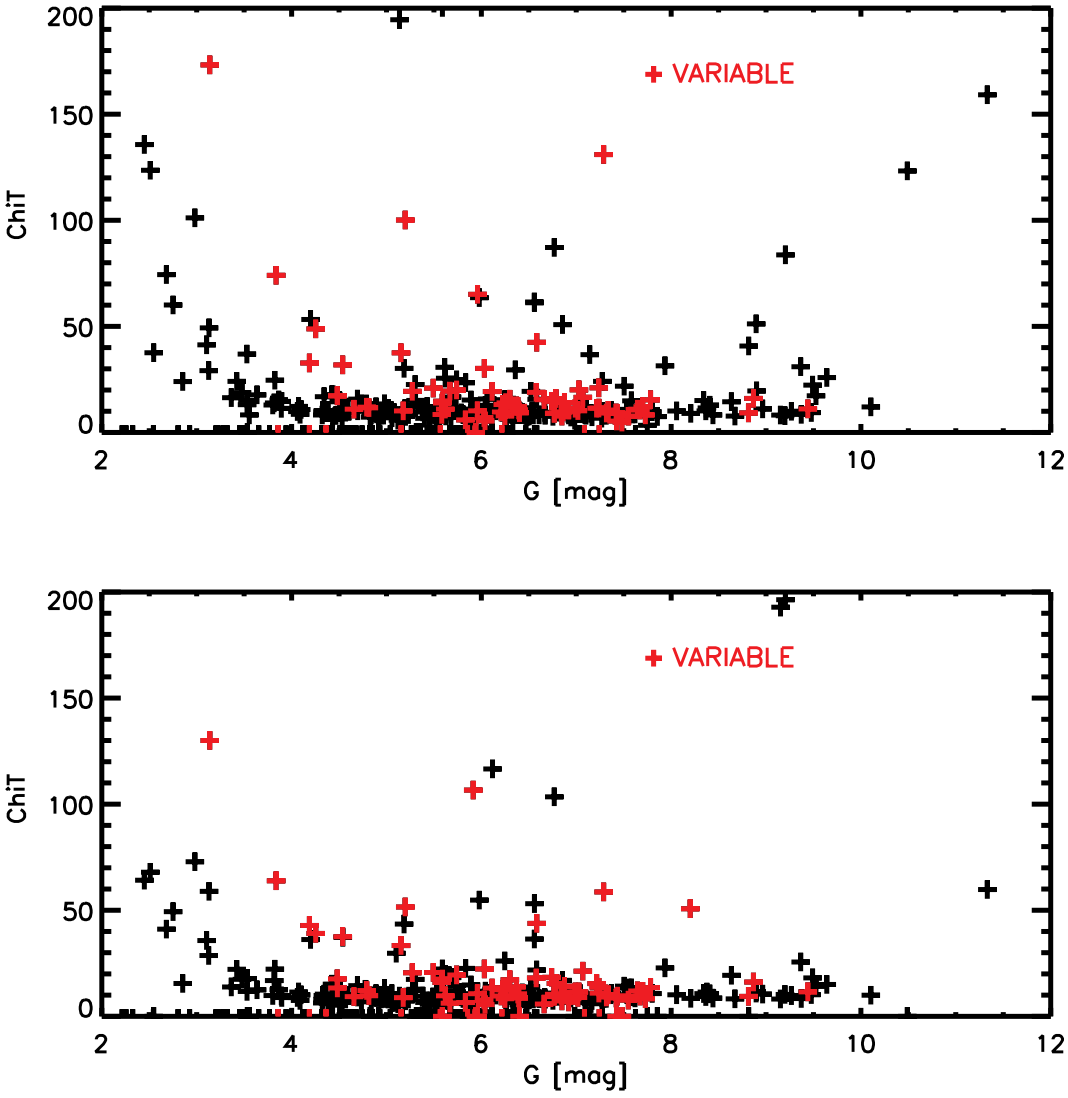

Fig. 7. UVES_POP spectra: *Upper panel:* ChiT versus Gmag of the XSL stars. *Lower panel:* An adjusted ChiT, i.e. a ChiT runned after a small rescaling of the XSL spectrum.

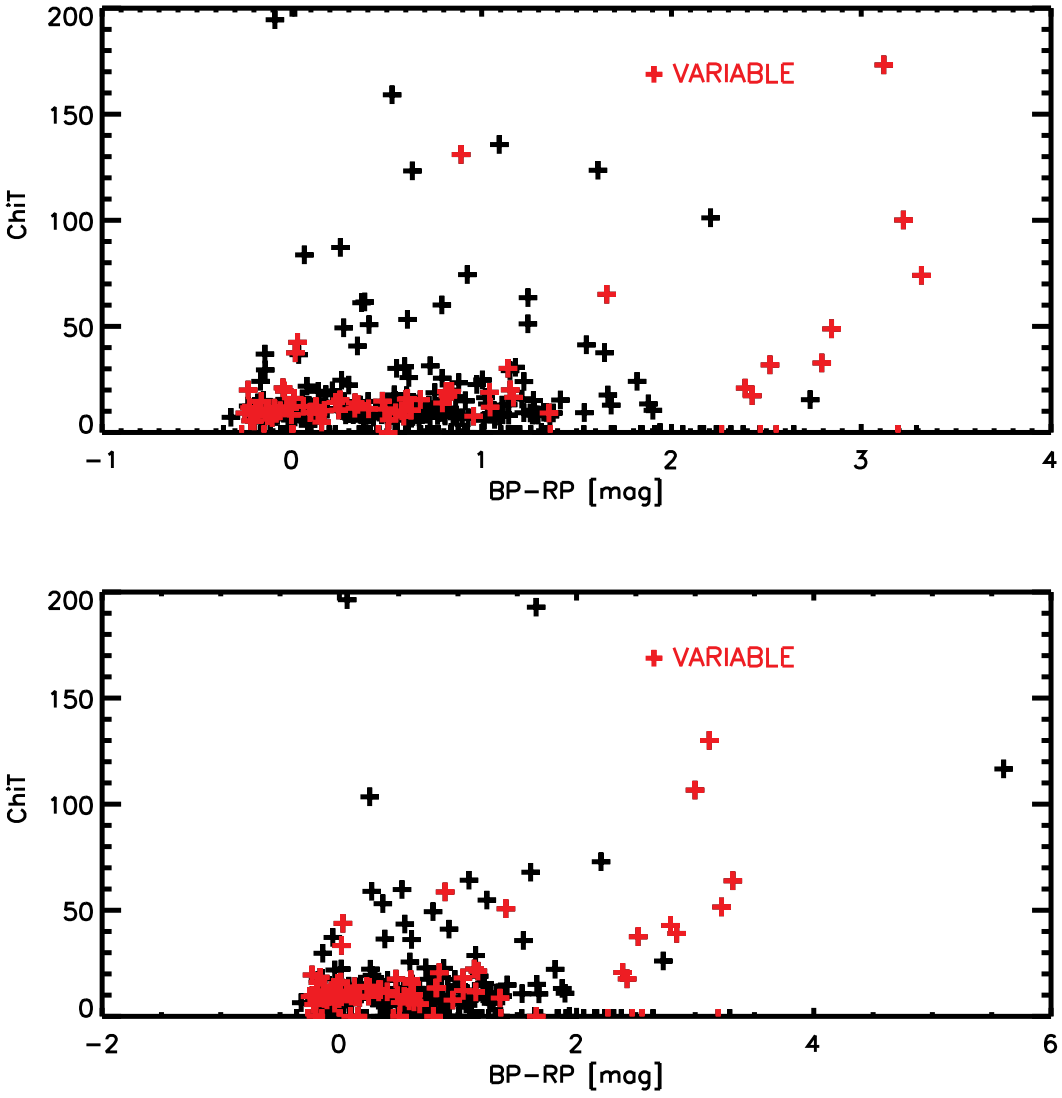

Fig. 8. UVES_POP spectra: *Upper panel:* ChiT versus BP-RP mag of the XSL stars. *Lower panel:* An adjusted ChiT, i.e. a ChiT runned after a small rescaling of the XSL spectrum.

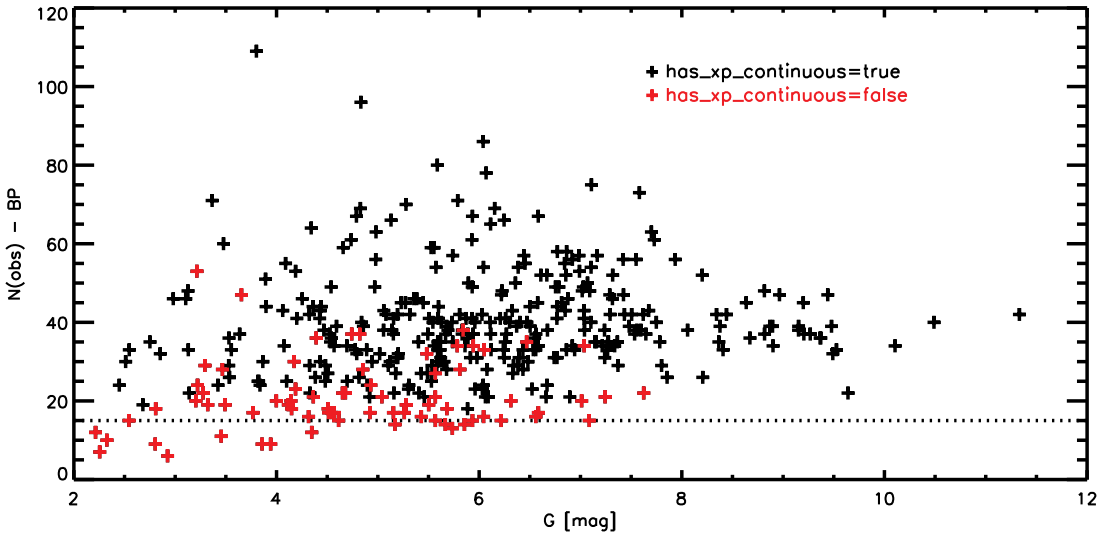

Fig. 9. UVES_POP spectra: Number of observation in BP-band versus Gmag.

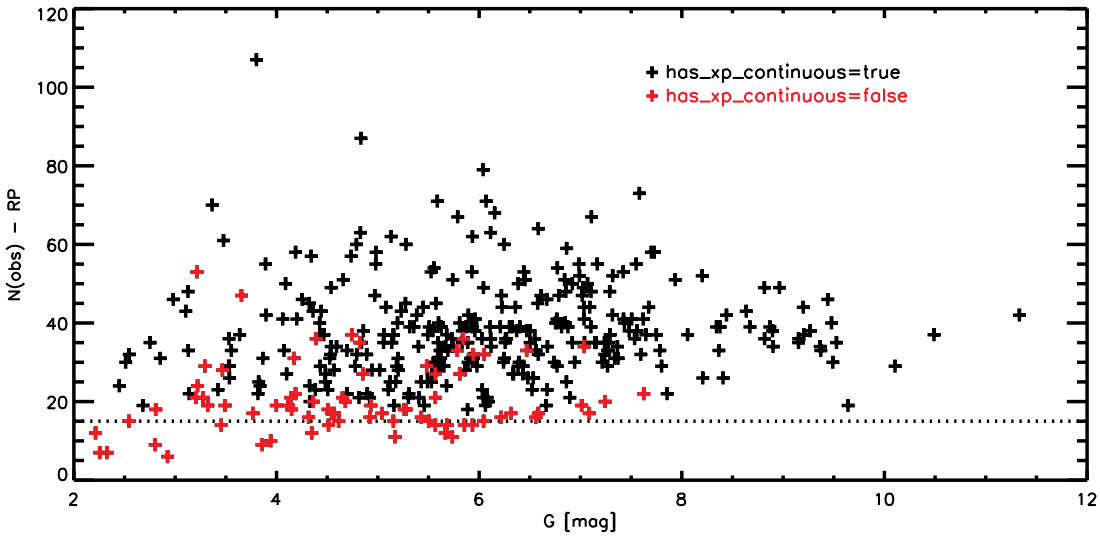

Fig. 10. UVES_POP spectra: Number of observation in RP-band versus Gmag.

Table 4. UVES_POP library. New identified Gaia-DR3 matches.

Name	Gaia-ID paper	Gaia DR3 ID re-retrieved	2MASS-ID
Castor	none	892348694913501952	07343598+3153184
HD020010	none	5059348952161258624	03120443-2859156
HD036646	none	3217181124645104384	none
HD066811	none	5534788672055388032	08033506-4000112
HD080404	none	5300300156538723328	09170540-5916308
HD083368	none	5410092611674819840	09362541-4845042
HD105113	none	3466217935643483392	12060519-3257402
HD120709	none	6170485544575679104	13514960-3259387
HD145792	none	6049891517263095168	16134549-2425196
HD154873	none	5950941488064653056	17102084-4644182
HD162306	none	4040709497081178496	17515522-3504577
HD162587	none	4040799485267855232	17532345-3453424
IC2391_47	none	5318309534208474112	08461506-5307444

Table 5. 302 matches between UVES_POP spectra and BR/RP spectra.

Gaia DR3 ID	Name	chiT	mean_flux(BP/RP)/mean_Flux(UVES_POP) between 700 and 750 nm
4976894960284258048	HD000142	8.27	0.98
2309813109479463040	HD000739	5.74	1.02
4900108950849461248	HD001581	12.28	1.01
4683897617110115200	HD002151	74.40	0.79
4925088171405735680	HD003158	7.48	1.02
4988805179472314368	HD007312	7.93	1.03
4934923028038871296	HD007570	7.21	NaN
4911306239828325760	HD010361	7.76	1.00
4912580642524184960	HD010647	9.63	1.00
4712976435488313216	HD010840	8.95	1.05
2461043306017773696	HD011171	9.54	1.03
4961654629610389376	HD011643	9.18	1.00
2492286341237939328	HD012642	0.00	NaN
5119953517683718400	HD014802	11.29	1.03
4748477123328077696	HD018242	101887.07	0.00
5065109121778224768	HD018466	10.20	0.99
5073246882211822720	HD018692	12.50	0.98
5059348952161258624	HD020010	12.94	0.98
4727337637895749888	HD020234	31.83	0.92
4722135642226902656	HD020807	9.36	1.02
4860808694421129600	HD021423	10.02	1.01
5084391257193086336	HD024616	8.52	0.99
4840555106121661952	HD026612	11.95	0.99
4676528209144882304	HD027256	29.13	0.98
4897544649215563008	HD027411	12.79	0.96
4615219578218188928	HD029138	8.94	1.04
3188395880157173120	HD030562	25.57	1.11
3289656915346252032	HD030677	9.51	1.00
2960561059245715968	HD032667	10.30	1.01
3212075924646082176	HD033256	10.25	1.03
3215512718819788288	HD034055	100.11	0.64
2954791234539940864	HD034295	9.08	0.98
4772547941122028672	HD035072	15.11	1.03
3217181124645104384	HD036646	14.64	0.88
3337991583943473280	HD036861	36.95	0.92
3017187866581306368	HD036959	7.11	1.03
3215970150016618880	HD037055	10.54	1.02
4766376347995853056	HD037227	8.16	0.99
3224479171721217280	HD037490	7.95	0.99
2907306935108521088	HD037495	8.10	0.97
2901273846086933888	HD037811	7.54	0.99
2964000159821218432	HD038393	18.75	1.03
4792774797545800832	HD039060	24.63	0.89
2995725777563537792	HD040136	17.70	1.05
5572962581890940160	HD042682	17.34	0.95
3006953165675574272	HD045725	282.38	0.53
2952064789303651200	HD046185	9.13	1.00
5580297801916850048	HD046462	10.98	1.01
2923818473022846976	HD047116	12.11	1.00
5501284972198817536	HD047306	9.77	1.05

Table 6. 302 matches between UVES_POP spectra and BR/RP spectra.

Gaia DR3 ID	Name	chiT	mean_flux(BP/RP)/mean_Flux(UVES_POP) between 700 and 750 nm
5584339709742678528	HD049131	19.98	1.05
2926942116843379968	HD049333	5.49	1.00
2952758856018716288	HD049888	24.09	NaN
2922367976673391232	HD050896	42.47	0.95
2947766145514856064	HD051876	10.75	NaN
5509770796785820416	HD055892	10.26	1.04
3030977013710528768	HD057167	10.99	1.04
3028075299446506880	HD058343	37.54	1.03
5605851482951097600	HD058377	7.55	1.00
5618635367207942400	HD058978	7.14	1.00
5493588665684618752	HD059468	8.24	1.00
5586381090520538880	HD059967	7.70	1.00
5591877995828084992	HD060498	8.03	0.97
5614158258941878144	HD061429	10.98	1.07
5599824647403389952	HD062623	14.64	1.07
5487758058599409920	HD062714	11.49	0.98
5588607120530408832	HD063077	11.19	1.03
5600487068801314048	HD064972	7.37	1.01
5714859436545926016	HD065810	10.85	1.03
5537802295990019968	HD065925	30.20	1.19
5597444857563449600	HD066698	8.25	1.00
5698015743046182272	HD067523	60.07	0.92
5541506859905246976	HD068761	8.32	1.00
5519380081746387328	HD068808	13.87	NaN
3065006795502406528	HD070574	8.81	1.01
5523058154294668544	HD072067	12.48	1.07
5523539671667837952	HD073634	10.80	1.03
5521599549054493696	HD074180	8.28	NaN
5329412200783621120	HD074272	10.19	1.03
5625409698860914432	HD074966	5.38	NaN
5329941409468845824	HD075063	15.30	1.09
5329890866290156288	HD075276	0.00	NaN
5316688377342006912	HD076131	9.38	0.97
5524337882746365952	HD076341	10.49	1.04
5730371484020807808	HD076932	10.66	0.97
5328473355278379136	HD077020	30.69	0.88
5218900665015560064	HD078791	10.36	1.02
5300463124771216896	HD079698	0.00	NaN
5423801425726635264	HD080057	0.00	NaN
5313280715934982144	HD081471	0.00	NaN
5306951892646516992	HD082668	37.57	NaN
3854247614992959872	HD083240	15.72	0.97
5247689452847083264	HD083312	21.79	0.92
5410092611674819840	HD083368	9.40	1.03
5309205822766572032	HD084461	9.67	1.03
615943806835727872	HD084937	8.97	0.99
5432365968466188800	HD085552	14.91	0.95
5253294179046079872	HD088366	739.29	0.21
5358432740734758528	HD089587	9.68	0.99
5365315790907090944	HD089736	13.46	0.98

Table 7. 302 matches between UVES_POP spectra and BR/RP spectra.

Gaia DR3 ID	Name	chiT	mean_flux(BP/RP)/mean_Flux(UVES_POP) between 700 and 750 nm
5441977138905560576	HD090132	12.95	0.97
5255045082580350080	HD090177	619.40	0.31
5351701122540337792	HD090772	17.20	1.07
5440991156144997504	HD091793	32.73	0.85
5199386085968886528	HD092305	24.08	1.03
5364553966782362752	HD092626	9.51	0.98
5350370026290390912	HD092740	13.97	1.02
5254268071479968512	HD093131	12.24	0.98
5388486280466017536	HD094660	9.02	1.00
5338220285385672064	HD094910	131.01	0.70
5241879697825137280	HD094963	8.07	1.01
5338040205991255296	HD095950	20.88	1.00
5340725660047993600	HD096917	10.26	1.01
5339394082770287232	HD097082	18.88	1.04
5337669189536576000	HD097253	10.25	0.98
5337672556798367360	HD097534	0.00	NaN
5398114596413589120	HD099322	8.16	0.99
3478127463341507072	HD100623	24.75	1.14
5333974693006467328	HD100841	49.31	1.09
5372587514128271232	HD101065	31.40	0.90
5343168568718268800	HD101584	10.48	0.98
5333358794709001216	HD101947	11.90	1.02
5334608080460780672	HD102878	14.72	1.09
5334558087036932992	HD104111	0.00	NaN
5836666564476158336	HD104237	61.49	0.75
3575733210781078016	HD104304	0.00	NaN
5856117406232328832	HD105056	9.11	1.00
6052948267023660032	HD105071	13.18	0.97
3466217935643483392	HD105113	7.97	1.01
6053245135154609280	HD105211	9.83	1.02
6054416870941383296	HD106068	5.93	1.01
6058271964884299520	HD107446	41.30	0.90
6058439910929477120	HD108968	19.35	0.97
5842864786417716608	HD109026	13.90	1.07
3501215734352781440	HD109379	135.70	0.67
5856874901058844160	HD109867	9.62	1.00
3516934936698212352	HD109931	7.18	1.00
3515784744456187520	HD110385	9.03	1.05
6127817277882986496	HD110458	9.28	1.00
3495820568233709952	HD111786	8.52	1.00
5861887402770403328	HD112272	20.08	0.93
6081364732673284864	HD112409	194.54	0.58
6055990993633719040	HD112842	5.74	NaN
5858915766471945984	HD113904	200.28	6.17
6080068545902295040	HD114365	6.97	1.02
3511640513332581504	HD114642	8.71	1.01
6062166950467100032	HD114837	13.41	0.96
5859173945562774272	HD115363	7.62	1.00
3507879565090229888	HD115617	6.84	1.01
5869816805762242560	HD115778	0.00	NaN

Table 8. 302 matches between UVES_POP spectra and BR/RP spectra.

Gaia DR3 ID	Name	chiT	mean_flux(BP/RP)/mean_Flux(UVES_POP) between 700 and 750 nm
6063901120782965248	HD115842	8.30	1.00
6161403234933741312	HD117440	0.00	NaN
3712858597278688000	HD118022	12.09	1.03
6070423698625106560	HD118767	74.05	0.96
6095384532415991296	HD121416	8.13	1.01
6089748096519831296	HD123515	10.19	1.01
6092862428844018176	HD125809	9.16	0.99
1284257623084916480	HD128167	16.87	1.09
5877902682962174208	HD131217	15.44	0.93
5900360581589586432	HD133518	29.48	NaN
5799008222524417408	HD134453	48.81	0.79
5876804064701201792	HD136239	9.17	0.97
5902781985395901056	HD136351	9.15	0.99
5883156699265571968	HD136537	9.41	0.97
5795787581164356480	HD137509	9.41	1.02
5888915425810310144	HD137753	8.14	0.99
6009538585839374336	HD140901	23.43	1.10
5835544375510084352	HD142758	11.36	0.99
5832957602573458304	HD143448	21.03	0.95
5998513610949980544	HD143928	8.35	1.00
6042226413691523712	HD145482	10.65	1.06
5821125860988362752	HD145689	10.96	1.00
6049891517263095168	HD145792	9.30	NaN
5989870075180588160	HD145842	349.11	0.38
5935484313053503360	HD146143	13.44	1.04
6037514800199297152	HD146836	6.64	1.00
4323764712422936704	HD148184	53.25	1.40
5830307779535002624	HD148291	7.83	NaN
5942196311827421568	HD148379	6.89	0.99
5765239059323778560	HD148451	12.88	1.03
5968761685316464128	HD148688	10.14	0.96
6052319827355147264	HD148898	17.82	0.95
5941151986262863104	HD148937	8.26	1.00
4358031335898505472	HD149661	5.70	1.00
5968019618048049664	HD151932	15.68	0.97
5966519162610481664	HD152003	7.69	0.98
5966501501704611840	HD152235	19.29	0.92
5966509541883497984	HD152270	61.13	0.95
4127420626087054976	HD152311	16.68	0.97
6034500386022688896	HD152429	8.35	0.99
5917537534527580160	HD152786	123.56	0.67
4111834567779557376	HD154365	16.56	1.06
5950908193476521856	HD154811	15.96	0.96
5950941488064653056	HD154873	50.83	0.79
5973552497981951360	HD155416	8.40	0.99
5979008205986483712	HD155806	19.49	0.94
4109034455276324608	HD156026	15.28	0.99
5979460655025099776	HD156098	10.21	NaN
5951165616611763456	HD156274	22.57	0.94
5952743072247166464	HD156385	87.19	1.22

Table 9. 302 matches between UVES_POP spectra and BR/RP spectra.

Gaia DR3 ID	Name	chiT	mean_flux(BP/RP)/mean_Flux(UVES_POP) between 700 and 750 nm
5952663872981837440	HD156575	9.89	0.99
5972810563833563648	HD157038	30.19	0.92
4053887521876855808	HD160529	65.11	NaN
5945941905576552064	HD160691	8.21	1.01
4041462250189194752	HD162144	8.71	1.01
4041579137662906752	HD162187	22.29	0.98
4041561171865960448	HD162305	9.76	1.01
4040709497081178496	HD162306	16.05	0.92
4040810720855068288	HD162374	7.69	1.05
4041599414270468480	HD162391	10.47	1.01
4040804085044179968	HD162393	10.12	1.00
4041570174122317440	HD162514	40.73	NaN
4040792716343899520	HD162515	9.93	1.01
4040842056948543232	HD162576	7.23	1.02
4040814740947685376	HD162586	15.61	1.09
4040799485267855232	HD162587	63.52	1.11
4040838380456117888	HD162630	10.90	1.01
4040801031403875456	HD162631	10.33	0.99
4040825358109531520	HD162678	5.32	1.01
4040801890398044928	HD162679	36.67	NaN
4040800722166107136	HD162680	3.35	NaN
4063703377509158656	HD162719	14.13	0.99
4040825529908227072	HD162724	15.80	1.05
4040824258597614976	HD162725	12.25	0.97
4040826526340512128	HD162780	9.91	1.00
4040821200580732800	HD162781	4.86	NaN
4040847760668408832	HD162817	8.74	1.02
4040834635242511616	HD162888	8.24	1.03
4040822811103190272	HD162889	10.15	0.99
4063417332766504576	HD163318	11.55	0.99
4037358426191922688	HD163652	8.93	0.99
4035245096041719424	HD163745	7.43	1.03
4037463292131089536	HD163758	6.68	1.06
4069523951550440832	HD163800	10.00	1.00
4177224620176470912	HD164259	15.13	1.07
4066022591147527552	HD164794	7.83	1.01
4076939607797219968	HD170235	19.36	1.13
4093346863913375232	HD171012	15.45	1.14
6444391655040244992	HD189124	173.22	0.87
6427464123776727168	HD190248	16.39	1.05
6850594406170094080	HD191301	10.48	0.99
6674382927491854848	HD196171	23.99	0.85
6777965134807570560	HD198357	9.31	0.97
6476205852190128512	HD199642	10.42	1.00
6881531326185815808	HD199728	7.82	1.01
6781898465857257472	HD199951	6.09	1.01
6832163911228998784	HD202320	8.24	1.02
6401464693867773568	HD203608	8.54	1.03
6894198082390222464	HD203875	9.91	1.02
6828125061423674752	HD204075	13.11	1.01

Table 10. 302 matches between UVES_POP spectra and BR/RP spectra.

Gaia DR3 ID	Name	chiT	mean_flux(BP/RP)/mean_Flux(UVES_POP) between 700 and 750 nm
6412595290592307840	HD209100	8.23	1.03
6611527455503953536	HD210066	17.64	0.97
6611824083124944384	HD210111	8.33	1.01
6625981669722090112	HD210848	7.30	1.02
6508776375901968640	HD211415	8.38	0.98
6382933589314905472	HD211998	6.88	0.98
6520955322608285824	HD213080	101.14	0.79
6515731817739456256	HD216435	9.26	1.00
6492406743706907776	HD219571	11.31	1.03
4040808968507325312	HD320764	19.58	1.05
4040804158143154560	HD320768	8.17	1.01
4040844840088710656	HD320847	12.02	0.97
4040800893964862080	HD320865	25.89	0.95
4040820341586756224	HD320868	12.85	0.99
5317972984874048000	IC2391_1	10.91	1.01
5321180844402213504	IC2391_2	11.57	0.99
5321196301989898624	IC2391_4	30.98	0.97
5318049534067120512	IC2391_5	7.71	0.98
5318169866178702976	IC2391_8	8.80	1.02
5318565651704167680	IC2391_10	9.22	0.98
5321582715910437376	IC2391_11	8.25	1.01
5318113550566048256	IC2391_13	12.24	1.07
5318541982145960064	IC2391_16	7.27	1.02
5321626009181506688	IC2391_17	8.93	0.99
5318117330136955392	IC2391_18	11.42	1.08
5318083382715884928	IC2391_19	8.27	1.00
5318547273548061824	IC2391_20	15.09	1.05
5318543425255007872	IC2391_21	10.98	1.06
5318111351542618496	IC2391_22	15.06	0.95
5318554111133634432	IC2391_23	9.77	1.00
5318501368933313280	IC2391_26	51.17	0.86
5318575448532526848	IC2391_28	7.50	NaN
5318499822745556992	IC2391_29	10.28	1.01
5318485048061677952	IC2391_31	8.47	1.06
5317874475508122752	IC2391_33	14.46	1.11
5318486456810851072	IC2391_34	11.67	1.07
5317887321743547264	IC2391_35	9.94	0.97
5318579606060497408	IC2391_36	9.38	0.99
5318285173154004352	IC2391_40	15.35	0.96
5318318918726520576	IC2391_41	9.99	1.00
5318270608933694208	IC2391_42	11.08	0.99
5318215495911387136	IC2391_43	11.08	0.98
5318630973868090496	IC2391_44	7.86	0.97
5318327165064125696	IC2391_45	8.52	0.99
5318323935248518144	IC2391_46	8.68	1.00
5318309534208474112	IC2391_47	83.69	1.59
5318418939925144448	IC2391_49	10.56	0.98
5318412686452956800	IC2391_50	10.22	0.99
5318182197020586112	IC2391_52	17.28	0.95
4041572819821784192	NGC6475_8	123.25	NaN
4041565840444161536	NGC6475_10	159.16	2.06
4041561339318438912	NGC6475_22	258.11	7.56